

Complete curriculum document · NSQF-aligned · industry-validated

Anvi Innovate AI Academy Full End-to-End Curriculum

A complete, government-mappable and IT-industry-aligned AI/ML training curriculum taking learners from zero to deployment-ready across six specialisation tracks. Every subject, module, topic, lab, real-world work scenario, assessment instrument and hour allocation is specified.

Curriculum version 2026.1 · Prepared August 2026 · Mapped to QP **SSC/Q8113** (IT-ITeS SSC / NASSCOM, NSQF Level 7) and NSQF Model Curriculum v3.0 (480 h) · Extended to 960 h for industry readiness

1 · Programme identity & compliance summary

Programme name	Anvi Innovate AI Academy — Applied AI Professional Programme
Curriculum version	2026.1 (re-validated per cohort)
Anchor qualification	SSC/Q8113 — AI-Machine Learning Engineer , NSQF Level 7
Awarding body	IT-ITeS Sector Skill Council (NASSCOM)
NCO alignment	NCO-2015 / 2512
Sector / Sub-sector	IT-ITeS / Future Skills · Occupation: Artificial Intelligence & Big Data Analytics
Statutory core	480 h per NSQF Model Curriculum v3.0 (144 Theory + 246 Practical + 90 OJT)
Total programme	960 h (480 h statutory core fully embedded + 480 h industry extension)
Duration	24 weeks full-time @ 40 h/week · 44 weeks part-time @ ~22 h/week
Entry (academy)	Final-year student or graduate — Engineering / Technology / Statistics / Mathematics / Computer Science / allied; English reading comprehension; basic computer literacy
Pass standard	70% aggregate; ≥70% on every technical NOS module (matches QP rule)
Exit tracks	AI Engineer · GenAI/LLM Engineer · MLOps/LLMOps Engineer · Data Scientist · Data & RAG Engineer · AI Solutions Architect
Award	Academy certificate + NSQF-aligned assessment through SSC-approved assessment agency + role certification of record

Compliance actions before first cohort: (1) Confirm the currently live version of SSC/Q8113 and Model Curriculum v3.0 with IT-ITeS SSC — the published QP v2.0 carries review/deactivation dating of 2025 and MC v3.0 shows validity to 22/09/2025. (2) Q8113 nominally expects a graduate with 2–3 years' experience; for pure freshers, register learners additionally against a fresher-appropriate QP (e.g. *Certified AI Associate* or *AI Application Developer*, NSQF 4.5, NIELIT) and use Q8113 as the aspirational exit standard. (3) Engage an SSC-emplanelled assessment agency before enrolment.

2 · Governing standards — the compliance spine

The curriculum is built *around* the National Occupational Standards, not retrofitted to them. Every NOS performance criterion below is delivered by a named module in §5–§10 and assessed by a named instrument in §12.

2.1 Compulsory NOS and statutory weightage

NOS	Title	Theory	Practical	Total	QP weight	Delivered by
SSC/N8121	Evaluate technical performance of algorithmic models	30	70	100	25%	C02, C05, G04, all capstones
SSC/N8122	Develop software code to support deployment of algorithmic models	25	75	100	25%	C05, R3, all track projects
SSC/N9001	Manage your work to meet requirements	25	75	100	10%	P01, sprint cadence
SSC/N9002	Work effectively with colleagues	20	80	100	10%	P01, pair programming, code review
SSC/N9004	Provide data/information in standard formats	per SSC criteria			—	P01, documentation & reporting standard
SSC/N9006	Build and maintain relationships in a workplace	per SSC criteria			—	P01, team projects, stakeholder simulation
SSC/N9014	Maintain an inclusive, environmentally sustainable workplace	10 h	20 h	30 h	—	P02

2.2 SSC/N8121 — performance criteria → delivery map

PC	Performance criterion (verbatim)	Module	Assessed by
PC1	Define technical specifications and limitations of the system running the algorithmic model	C02 M1	Spec document + viva
PC2	Define the data flows and evaluate the data structures in the algorithmic model	C02 M1	Data-flow diagram assessment
PC3	Evaluate the speed, runtime and memory interdependencies of the algorithmic model	C02 M2	Profiling lab report
PC4	Evaluate the parallel programming constraints of the system	C02 M3	Parallelism practical
PC5	Modify and test the algorithmic model to fit speed, runtime and memory limitations	C02 M4	Optimisation practical (before/after metrics)
PC6	Update documentation on the algorithmic model	C02 M5	Model card review
PC7	Provide feedback on model performance to the team responsible for design/development	C02 M5	Structured feedback exercise

Knowledge & Understanding covered (KU1–KU15): system purpose and aims; organisational policy for evaluating model performance, data sharing and documentation; data sources and access; escalation and stakeholder involvement; standard templates and tools; interpreting technical specifications and system limits; algorithmic model families; data flows and structures; speed/runtime/memory dependency analysis; parallel computing architectures **SISD, SIMD, MISD, MIMD**; performance measurement methodology including **Big-O notation**; anomaly identification and referral; operating in **Linux/Ubuntu/Windows** environments. **Generic Skills (GS1–GS4):** deciding suitable courses of action; planning and organising own work to deadlines; referring anomalies to supervisor; checking work is complete and error-free.

2.3 SSC/N8122 — performance criteria → delivery map

PC	Performance criterion (verbatim)	Module	Assessed by
PC1	Define technical specifications and limitations of the autonomous system	C05 M1	SRS document

PC	Performance criterion (verbatim)	Module	Assessed by
PC2	Evaluate the design of data flows and structures in the autonomous system	C05 M1	HLD review
PC3	Evaluate the design of core algorithmic models in the autonomous system	C05 M2	Design critique
PC4	Evaluate the parallel programming constraints of the autonomous system	C05 M2	Constraint analysis
PC5	Integrate specifications, data flows, models and parallel constraints into reusable code	C05 M3	Code artefact + reuse review
PC6	Create documentation on developed software code for relevant stakeholders	C05 M3	Docs rubric
PC7	Validate developed code with relevant stakeholders	C05 M3	Walkthrough/demo
PC8	Create and execute unit test cases to analyse the performance of the code	C05 M4	Test suite + coverage
PC9	Evaluate the defects in the code and identify corrective actions	C05 M4	Defect log
PC10	Implement corrective actions to optimise code performance	C05 M4	Refactor practical
PC11	Submit tested and optimised code for approval by appropriate people	C05 M5	PR approval gate

Scope elements to be taught explicitly: Software Requirements Specification (functional and non-functional requirements); High-Level Design (technical solution and rationale, assumptions, constraints, dependencies, interfaces); Low-Level Design (programming structures, coding standards, checklists). **KU coverage (KU1–KU18)** includes approval processes for code designs, code-generation and unit-testing tool ranges, component reuse determination, defect taxonomy and how recorded corrective actions improve future design.

3 · Programme architecture & hour allocation

960 h

total programme
24 weeks full-time

480 h

statutory core embedded
Model Curriculum v3.0

~62%

practical & project hours
practical-dominant by design

90 h

mandatory OJT
industry immersion

3.1 Semester structure

Stage	Weeks	Subjects	Theory	Practical	Total	Gate
A · Foundations	1–6	F01–F08	90	150	240 h	Gate 1 — Foundations examination
B · Core AI	7–12	C01–C05	96	144	240 h	Gate 2 — NOS 8121/8122 practical
C · Applied GenAI	13–18	G01–G05	84	156	240 h	Gate 3 — RAG/agent system with evals
D · Specialisation	19–22	One of R1–R6	48	112	160 h	Gate 4 — deployed track project
E · Capstone	23–24	X01	10	70	80 h	Gate 5 — capstone defence
Parallel	1–24	P01, P02	34	56	90 h	Continuous evaluation
Post	25+	X02 OJT	—	—	90 h	Supervisor sign-off
TOTAL	24 wks	—	362	688	1050 h	incl. OJT

3.2 Statutory core embedding map

Every hour of the government Model Curriculum is accounted for inside the academy programme.

Model Curriculum module (v3.0)	Statutory hours	Delivered inside
Module 1 (Bridge): AI & Big Data Analytics — An Introduction	4 T + 8 P = 12	F07
Module 2 (Bridge): Product Engineering Basics	8 T + 18 P = 26	F08
Module 3 (Bridge): Basic Statistical Concepts	8 T + 16 P = 24	F04
Module 4 (Bridge): Development Tools and Usage	10 T + 18 P = 28	F01, F02, F06
Module 5: Performance Evaluation of Algorithmic Models (SSC/N8121)	40 T + 50 P = 90	C02
Module 6: Code Development (SSC/N8122)	40 T + 80 P = 120	C05
Module 12: Inclusive & environmentally sustainable workplaces (SSC/N9014)	10 T + 20 P = 30	P02
Employability Skills (DGT/VSQ/N0102), Modules 6–17	24 T + 36 P = 60	P01
On-the-Job Training (mandatory)	90	X02
Statutory total	144 T + 246 P + 90 OJT = 480 h	Fully embedded

4 · Subject catalogue

Code	Subject	Hours	Prerequisite	NOS
F01	Programming Foundations with Python	48	—	Bridge M4
F02	Software Engineering, Git & Collaboration	28	F01	Bridge M4, N9002
F03	Data Handling, Wrangling & Visualisation	36	F01	N9004
F04	Mathematics & Statistics for AI	40	—	Bridge M3
F05	Databases & SQL for AI Systems	28	F01	N9004
F06	Linux, Networking & Cloud Fundamentals	24	—	Bridge M4, KU15
F07	AI & Big Data Analytics — Industry Landscape	12	—	Bridge M1
F08	Product Engineering & SDLC Basics	24	—	Bridge M2
C01	Classical Machine Learning	56	F01,F03,F04	KU9
C02	Model Evaluation & Performance Engineering	90	C01	SSC/N8121
C03	Deep Learning	48	C01	KU9
C04	Transformers & Representation Learning	26	C03	KU9
C05	Software Code Development for Model Deployment	120	F02,C01	SSC/N8122
G01	Large Language Models & Prompt Engineering	44	C04	—
G02	Retrieval-Augmented Generation	56	G01,F05	—
G03	AI Agents & Tool Orchestration	48	G02	—
G04	LLM Evaluation, Observability & Cost	44	G02	SSC/N8121
G05	Responsible AI, Security & Data Governance	48	G01	N9014
R1–R6	Role specialisation track (one)	160	Gate 3	N8121, N8122
P01	Professional Practice & Employability	60	—	N9001,N9002,N9004,N9006
P02	Inclusive & Sustainable Workplace	30	—	SSC/N9014
X01	Capstone Project	80	Gate 4	All
X02	On-the-Job Training / Industry Immersion	90	Gate 5	All

5 · Stage A — Foundations (Weeks 1–6 · 240 h)

F01 · Programming Foundations with Python

48 h · 16 T / 32 P

Prerequisite None

Outcome Writes tested, modular, idiomatic Python for data and AI workloads

Modules & topics

M	Topics	Lab	h
M1	Interpreter, environments (<code>venv</code> / <code>uv</code>), package management, project layout, PEP 8, editors and debuggers	Provision a reproducible project skeleton with pinned dependencies	6
M2	Types, strings, f-strings, collections (list/tuple/dict/set), comprehensions, slicing, mutability semantics	Data-transformation kata set (12 problems)	8
M3	Control flow, functions, arguments/defaults, scope, closures, decorators, generators, iterators, context managers	Build a retry decorator and a streaming file reader	8
M4	OOP: classes, inheritance, composition, dunder methods, dataclasses, protocols; module and package design	Design a plug-in class hierarchy for data loaders	8
M5	Errors and exceptions, custom exceptions, logging, defensive programming, type hints, <code>mypy</code>	Add typing and structured logging to an existing script	6
M6	File and data I/O: CSV, JSON, JSONL, Parquet, YAML; HTTP with <code>requests</code> ; pagination, retries, rate limits	Build a resilient API-ingest client with backoff	6
M7	Testing with <code>pytest</code> : unit tests, fixtures, parametrisation, mocking, coverage; linting with <code>ruff</code>	Achieve ≥80% coverage on the ingest client	6

Real work scenario — You inherit an undocumented data-pull script that fails silently every third night. Reproduce the failure, add structured logging and typed interfaces, write regression tests that capture the bug, fix it, and raise a pull request with a written root-cause note.

Assessment Auto-graded coding challenge (40) · lab portfolio rubric (40) · code-review participation (20)

F02 · Software Engineering, Git & Collaboration

28 h · 10 T / 18 P

M	Topics	Lab	h
M1	Version control model, repositories, commits, branching strategies (trunk-based, GitFlow), merge vs rebase, conflict resolution	Resolve a deliberately conflicted merge	7
M2	Collaboration: pull requests, review etiquette, commit hygiene, semantic versioning, changelogs, issue tracking	Two-person PR exchange with mandatory review comments	7
M3	Coding standards, checklists, low-level design conventions, docstrings, README and architecture documentation	Author a repository standard and enforce it via linting	7
M4	CI basics: GitHub Actions, automated test and lint gates, branch protection, secrets handling	CI pipeline that blocks merge on test or lint failure	7

Real work scenario — A teammate's PR breaks the build and mixes three unrelated changes. Review it professionally, request specific changes, help split the commits, and get the pipeline green without a direct push to main.

NOS Supports SSC/N9002 (PC1–PC6), SSC/N8122 KU7–KU8

Assessment PR review rubric · CI pipeline practical

F03 · Data Handling, Wrangling & Visualisation

36 h · 12 T / 24 P

M	Topics	Lab	h
M1	NumPy: ndarray, dtypes, broadcasting, vectorisation, axis semantics, memory layout, random number generation	Replace loops with vectorised operations; benchmark speed-up	8
M2	Pandas: Series/DataFrame, indexing, selection, joins/merges, group-by aggregation, reshaping, time series, categoricals	Join four heterogeneous sources into one analysis table	10
M3	Data quality: missing values, duplicates, outliers, type coercion, encoding issues, validation rules, data contracts	Build a validation layer that quarantines bad rows	8
M4	Exploratory data analysis workflow; visualisation with Matplotlib and Seaborn; chart selection; communicating uncertainty	Produce a decision-oriented EDA report	10

Real work scenario — Business reports a revenue dashboard is "wrong". Trace the discrepancy through the join logic, discover duplicate keys inflating totals, quantify the impact, correct the pipeline, and write the incident summary for a non-technical stakeholder.

NOS SSC/N9004 (PC1–PC6) · SSC/N8121 KU3, KU10

F04 · Mathematics & Statistics for AI

40 h · 18 T / 22 P — embeds statutory Bridge Module 3

M	Topics	Lab	h
M1	Linear algebra: vectors, matrices, matrix multiplication, transpose, inverse, identity, rank, eigenvalues and eigenvectors, norms, dot product and cosine similarity	Implement matrix operations and cosine similarity from first principles	10
M2	Calculus for optimisation: derivatives, partial derivatives, chain rule, gradients, convexity, gradient descent and variants	Implement batch and stochastic gradient descent; visualise convergence	8
M3	Descriptive statistics: measures of central tendency (mean, median, mode), dispersion, variance, standard deviation, skew, percentiles; statistical anomalies — missing values, bias, outliers	Statistical profile of a raw production dataset	8
M4	Probability: sample spaces, conditional probability, Bayes' theorem, independence, distributions (normal, binomial, Poisson), CLT	Bayesian update exercise on a classification prior	7
M5	Inferential statistics: sampling, confidence intervals, hypothesis testing, p-values, type I/II error, statistical power, correlation vs causation	Design and analyse a controlled comparison	7

Real work scenario — A stakeholder claims a new feature "lifted conversion by 12%". Determine whether the sample supports the claim, compute the confidence interval, identify the confound, and present a defensible verdict.

F05 · Databases & SQL for AI Systems

28 h · 10 T / 18 P

M	Topics	Lab	h
M1	Relational model, keys, normalisation, schema design, data types, constraints, indexing fundamentals	Design a normalised schema for a transactional domain	7
M2	SQL: SELECT, filtering, joins (inner/outer/self), aggregation, GROUP BY/HAVING, subqueries, CTEs, window functions	Analytical query set of 20 graded problems	9
M3	Query performance: execution plans, index selection, cardinality, partitioning, avoiding N+1 access patterns	Reduce a slow query's runtime by an order of magnitude	6
M4	Access from Python (<code>SQLAlchemy</code> , drivers, connection pooling); transactions and isolation; NoSQL and vector-store preview	Parameterised, pooled data-access layer	6

Real work scenario — A nightly aggregation job has grown from 4 minutes to 90 minutes. Read the execution plan, identify the missing composite index and the accidental cross join, fix both, and document the before/after.

F06 · Linux, Networking & Cloud Fundamentals

24 h · 8 T / 16 P — satisfies SSC/N8121 KU15

M	Topics	Lab	h
M1	Linux/Ubuntu: filesystem, permissions, processes, package managers, shell scripting, SSH, cron, systemd basics; working across Linux and Windows	Automate a scheduled data job on a Linux VM	8
M2	Networking: TCP/IP, DNS, HTTP/HTTPS, TLS, ports, firewalls, proxies, REST semantics, status codes, latency vs throughput	Diagnose a failing service call end to end	6
M3	Cloud fundamentals: regions and availability zones, IAM and least privilege, compute, object storage, managed databases, secrets management, cost model	Provision least-privilege storage and compute for a workload	6
M4	Containers introduction: images, layers, Dockerfile, registries, volumes, port mapping	Containerise a Python service	4

Real work scenario — A model service works locally but returns 502 in the cloud. Work down the stack — container port binding, security group, IAM permission, secret injection — and produce a runbook so the next engineer resolves it in minutes.

F07 · AI & Big Data Analytics — Industry Landscape

12 h · 4 T / 8 P — statutory Bridge Module 1

M	Topics	Lab	h
M1	AI, ML, deep learning and generative AI — definitions and boundaries; the AI project lifecycle; data-to-value chain	Map a business problem to an AI solution pattern	4
M2	Industry use cases across BFSI, healthcare, retail, manufacturing, telecom and public services; build-vs-buy; failure patterns	Use-case teardown of a deployed industry system	4
M3	Occupational roles in AI & Big Data Analytics: AI engineer, ML engineer, data scientist, data engineer, MLOps, solutions architect, annotator — responsibilities, interfaces and career ladders	Role-mapping exercise against live job descriptions	4

M	Topics	Lab	h
M1	SDLC models; Agile and Scrum: sprints, backlog, estimation, stand-ups, retrospectives; Kanban and WIP limits	Run a two-week simulated sprint with ceremonies	7
M2	Requirements engineering: eliciting requirements, user stories, acceptance criteria, functional vs non-functional requirements, traceability	Convert an ambiguous business ask into a written SRS	7
M3	Design documentation: High-Level Design (solution rationale, assumptions, constraints, dependencies, interfaces) and Low-Level Design (structures, standards, checklists)	Produce HLD and LLD for a model-serving feature	6
M4	Delivery: release planning, timelines, dependency management, risk registers, definition of done, stakeholder communication	Build a delivery plan with a risk register	4

Real work scenario — The client changes a requirement mid-sprint. Assess the impact on scope, timeline and dependencies, update the SRS and risk register, and communicate the revised commitment in writing.

Gate 1 — Foundations Examination (Week 6): 90-minute closed-book theory paper (Python, statistics, SQL, Linux, SDLC) + 3-hour practical (ingest → clean → analyse → visualise → commit via PR) + repository review. Pass ≥70%.

6 · Stage B — Core AI (Weeks 7–12 · 240 h)

C01 · Classical Machine Learning

56 h · 20 T / 36 P

M	Topics	Lab	h
M1	Learning paradigms: supervised, unsupervised, semi-supervised, reinforcement; problem framing; train/validation/test discipline; overfitting and underfitting; bias–variance decomposition	Construct leakage-free splits for a temporal dataset	8
M2	Regression: linear, polynomial, regularisation (Ridge, Lasso, ElasticNet); assumptions and diagnostics; residual analysis	Regularisation sweep with learning curves	8
M3	Classification: logistic regression, k-NN, naïve Bayes, support vector machines and kernels, decision boundaries, class imbalance handling	Imbalanced classification with resampling and thresholds	10
M4	Trees and ensembles: decision trees, bagging, random forests, boosting (gradient boosting, XGBoost, LightGBM), stacking, feature importance	Tune a gradient-boosted model against a baseline	10
M5	Unsupervised learning: k-means, hierarchical clustering, DBSCAN, silhouette analysis, PCA and dimensionality reduction, anomaly detection	Segment customers and justify cluster count	10
M6	Feature engineering: encoding, scaling, binning, interactions, temporal features, target leakage, pipelines and column transformers, hyperparameter search	End-to-end reproducible pipeline object	10

Real work scenario — Your model scores 0.94 AUC offline and fails in production. Discover that a feature was computed using post-event information, rebuild the feature set with a strict point-in-time join, and requantify honest performance.

C02 · Model Evaluation & Performance Engineering STATUTORY

90 h · 40 T / 50 P — SSC/N8121

Delivers Model Curriculum Module 5 in full. This subject carries **25% of the QP assessment weight** and must be passed at $\geq 70\%$ independently.

M	Topics	Lab	PC	h
M1	Define model parameters. Technical specifications and limitations of the host system: CPU/GPU architecture, cores, clock, cache hierarchy, RAM and VRAM ceilings, storage I/O, OS constraints. Data flows and data structures inside the model: tensors, batches, feature stores, serialisation formats, precision (fp32/fp16/int8). Interface contracts and schema definition	Author a formal specification sheet documenting host limits and the model's data-flow graph	PC1, PC2	18
M2	Runtime, speed and memory analysis. Algorithmic complexity and Big-O notation ; time and space trade-offs; profiling (<code>cProfile</code> , <code>line_profiler</code> , <code>memory_profiler</code> , PyTorch profiler); latency percentiles (p50/p95/p99), throughput, cold start; memory leaks, fragmentation, garbage collection; batch-size versus memory curves	Profile a model end to end; produce a flame graph and a bottleneck report ranked by impact	PC3	20
M3	Parallel programming constraints. Flynn's taxonomy — SISD, SIMD, MISD, MIMD ; threads versus processes; the Python GIL; multiprocessing, vectorisation, BLAS threading; GPU execution model, kernels, occupancy, memory bandwidth; data parallel versus model parallel; Amdahl's law; synchronisation, race conditions and deadlock	Parallelise an inference workload; measure scaling efficiency against Amdahl's prediction	PC4	18
M4	Optimise to fit constraints. Quantisation (post-training and quantisation-aware), pruning, knowledge distillation, operator fusion, graph optimisation, ONNX export, batching strategies, caching, mixed precision; accuracy-versus-resource trade-off analysis; regression testing after optimisation	Reduce model latency and memory to a stated budget while holding accuracy within tolerance; present the trade-off curve	PC5	20
M5	Document and feed back. Model cards, evaluation reports, versioned documentation, reproducibility metadata; anomaly identification and referral; structured performance feedback to the design and development team; organisational policy for documentation and data sharing; escalation paths	Publish a model card and deliver a formal performance-feedback review to the modelling team	PC6, PC7	14

Real work scenario — Production inference must run within 200 ms at p95 on a 4-core CPU instance with 8 GB RAM, but the current model averages 850 ms. Establish the baseline, profile to locate the bottleneck, apply quantisation and batching, verify accuracy remains within the agreed tolerance, document the change in the model card, and formally report the trade-off to the data-science team.

Assessment Theory 30 marks · Practical 70 marks, mirroring the statutory split. Practical is graded on measured before/after evidence, not narrative.

C03 · Deep Learning

48 h · 16 T / 32 P

M	Topics	Lab	h
M1	Neurons, activation functions, feed-forward networks, loss functions, backpropagation derivation, computational graphs, autograd	Implement a 2-layer network and backprop manually, then in PyTorch	10
M2	Training dynamics: initialisation, optimisers (SGD, momentum, RMSProp, Adam, AdamW), learning-rate schedules, batch size, gradient clipping, vanishing/exploding gradients	Diagnose and repair a non-converging training run	10
M3	Regularisation and generalisation: dropout, batch/layer normalisation, weight decay, early stopping, data augmentation, transfer learning and fine-tuning	Transfer-learn a pretrained backbone to a small dataset	10
M4	Convolutional networks: convolution, pooling, stride/padding, receptive field, classic architectures, image pipelines	Train and evaluate an image classifier with augmentation	10
M5	Sequence models: RNN, LSTM, GRU, sequence-to-sequence, teacher forcing, limitations that motivate attention	Sequence model on time-series or text; compare to a transformer baseline	8

C04 · Transformers & Representation Learning

26 h · 10 T / 16 P

M	Topics	Lab	h
M1	Attention mechanism: query-key-value, scaled dot-product attention and the $\sqrt{d}$ scaling rationale, softmax weighting, multi-head attention, self versus cross attention, causal masking	Implement scaled dot-product and multi-head attention from scratch	10
M2	Transformer architecture: encoder and decoder stacks, residual connections, layer normalisation, feed-forward blocks, positional encoding (absolute, learned, rotary), quadratic complexity in sequence length	Assemble a minimal transformer block and verify shapes and gradients	8
M3	Representation learning: tokenisation (BPE, WordPiece, SentencePiece), vocabulary, embeddings, contextual versus static embeddings, embedding geometry, cosine similarity, pretraining objectives (causal LM, masked LM)	Train a tokeniser; probe embedding similarity behaviour	8

C05 · Software Code Development for Model Deployment STATUTORY

120 h · 40 T / 80 P — SSC/N8122

Delivers Model Curriculum Module 6 in full. Carries **25% of QP assessment weight**; must be passed at $\geq 70\%$ independently.

M	Topics	Lab	PC	h
M1	Define requirements. Software Requirements Specification: functional and non-functional requirements (latency, throughput, availability, security, cost); technical specifications and limitations of the target system; evaluating the design of data flows and data structures; interface and schema contracts; traceability matrix	Produce a complete SRS and reviewed data-flow design for a model-serving system	PC1, PC2	22
M2	Evaluate model and concurrency design. Reviewing core algorithmic model design for deployability; High-Level Design — technical solution and rationale, assumptions, constraints, dependencies, interfaces; parallel programming constraints of the target system; concurrency models, thread safety, statelessness, idempotency	Write and defend an HLD including a concurrency and constraint analysis	PC3, PC4	22
M3	Convert specifications into reusable code. Low-Level Design: programming structures, coding standards, checklists; modular and reusable component design; determining component reuse suitability; serving frameworks (FastAPI), request/response schemas (Pydantic), configuration and secrets, dependency injection, packaging, containerisation; code-generation tooling; stakeholder documentation and validation walkthroughs	Implement a documented, containerised, schema-validated inference service and validate it with stakeholders	PC5, PC6, PC7	34
M4	Unit testing and defect management. Test design and the test pyramid; unit test case creation and execution; fixtures, mocking, parametrisation, coverage measurement; performance and load testing; defect taxonomy, triage, root-cause analysis, corrective action; recording corrective actions to improve future design; refactoring for performance	Build a test suite with coverage gate; log, classify and remediate seeded defects; optimise the hot path	PC8, PC9, PC10	28
M5	Approval and release. Organisational approval process for code designs; pull-request and sign-off workflow; release notes and versioning; deployment readiness checklist; handover documentation	Submit optimised, tested code through a formal approval gate to release	PC11	14

Real work scenario — Data science delivers a notebook model that must go live in three weeks under a 99.5% availability target. Write the SRS and HLD, refactor the notebook into a tested, containerised, schema-validated service, load-test it, remediate the defects found, document it, and take it through code review and release approval.

Assessment Theory 25 · Practical 75, mirroring the statutory split. Deliverables: SRS, HLD, LLD, running service, test suite with coverage report, defect log, approved PR.

Gate 2 (Week 12): combined NOS practical — profile and optimise a supplied model to a stated budget (N8121) and deliver it as a tested, documented, containerised service (N8122). Both assessed independently at $\geq 70\%$.

7 · Stage C — Applied Generative AI (Weeks 13–18 · 240 h)

G01 · Large Language Models & Prompt Engineering

44 h · 16 T / 28 P

M	Topics	Lab	h
M1	LLM landscape: pretraining, instruction tuning, alignment (RLHF/RLAIF), model families, proprietary versus open-weights, context windows, capability and limitation profiles, model selection criteria	Benchmark three models on one task for quality, latency and cost	8
M2	Inference mechanics: tokenisation and token accounting, sampling parameters (temperature, top-p, top-k), stop sequences, determinism, max tokens, streaming, context-window management and truncation strategy	Token-budget calculator and sampling sensitivity study	8
M3	Prompt engineering: zero-shot, few-shot, chain-of-thought, role and system prompts, delimiters, output contracts, self-consistency, decomposition, prompt templating and versioning, anti-patterns	Build a versioned prompt library with regression cases	10
M4	Structured outputs and integration: JSON mode, schema enforcement with Pydantic , function/tool calling, validation and repair loops, error taxonomy, retries, idempotency, provider SDKs and gateways	Service returning strictly schema-valid structured output with repair	10
M5	Hallucination: causes, detection, grounding strategies, refusal design, confidence signalling, citation requirements	Measure and reduce unsupported-claim rate on a fixed task set	8

Real work scenario — A customer-facing summariser occasionally invents policy numbers. Build a reproducible failure set, add grounding and schema validation, define refusal behaviour, and demonstrate a measured reduction in fabricated fields.

G02 · Retrieval-Augmented Generation

56 h · 18 T / 38 P

M	Topics	Lab	h
M1	RAG architecture: why retrieval, indexing pipeline versus query pipeline, grounding and citation, failure modes, when RAG is the wrong tool	Baseline RAG implementation end to end	8
M2	Ingestion and parsing: document loaders, PDF/HTML/DOCX/table extraction, OCR, layout preservation, metadata extraction, incremental sync, deduplication, refresh scheduling	Ingest a heterogeneous corpus including scanned and tabular documents	10
M3	Chunking strategy: fixed, recursive, semantic, parent-document and hierarchical chunking; overlap; chunk size versus recall; metadata design for filtering	Controlled chunking experiment measured on retrieval metrics	10
M4	Embeddings and vector stores: embedding model selection and benchmarking, dimensionality, normalisation; index types (flat, HNSW, IVF), distance metrics, filtering, hybrid metadata queries, upserts, sharding and scale	Build and tune a vector index for recall and latency	12
M5	Retrieval quality: dense versus sparse (BM25), hybrid fusion, reranking (cross-encoder), query rewriting and expansion, multi-hop retrieval, top-k selection, context assembly and ordering	Raise recall@k and precision with hybrid search plus reranking	10
M6	Generation layer: grounded prompting, citation enforcement, context-window packing, conflicting-source handling, answerability and abstention	Citation-enforced answering with abstention on insufficient evidence	6

Real work scenario — An internal policy assistant answers confidently from outdated documents. Add document versioning and recency metadata, implement hybrid retrieval with reranking, enforce citations, and prove the improvement with retrieval and answer metrics on a fixed evaluation set.

G03 · AI Agents & Tool Orchestration

48 h · 16 T / 32 P

M	Topics	Lab	h
M1	Agent foundations: agent loop, reasoning and acting patterns, planning versus reactive execution, when an agent is unnecessary, cost and latency implications	Single-tool agent with explicit termination conditions	8
M2	Tool design: tool schemas and descriptions, function calling, argument validation, side-effect classification, idempotency, tool errors and recovery, Model Context Protocol servers	Build and expose a tool set including a stateful external action	10
M3	Orchestration frameworks: graph and state-machine execution, durable state and checkpointing, resumability, branching and conditional edges, subgraphs, multi-agent patterns (supervisor, hierarchical), handoffs	Stateful multi-step workflow that resumes after induced failure	12
M4	Memory: short-term context, conversation summarisation, long-term memory stores, retrieval-backed memory, memory hygiene and privacy	Add scoped, expiring memory to an assistant	8
M5	Control and safety: human-in-the-loop approval gates, permission scoping and least privilege for tools, loop and cost caps, timeouts, sandboxing, audit trails	Approval-gated agent with enforced spend and iteration ceilings	10

Real work scenario — An agent authorised to issue refunds must never act alone above a threshold. Implement permission scoping, a human approval gate, idempotent tool calls, cost caps and a complete audit trail; then demonstrate correct behaviour under retry and partial-failure conditions.

G04 · LLM Evaluation, Observability & Cost

44 h · 16 T / 28 P — reinforces SSC/N8121

M	Topics	Lab	h
M1	Evaluation strategy: what to measure, offline versus online, golden datasets, dataset curation and maintenance, annotation guidelines, inter-annotator agreement, test-set contamination	Construct a governed golden dataset with an annotation guide	8
M2	Retrieval metrics: recall@k, precision@k, MRR, NDCG, hit rate, context precision and recall — evaluated independently of generation	Retrieval-only evaluation harness with dashboards	8
M3	Generation metrics: faithfulness/groundedness, answer relevance, correctness, completeness, toxicity and safety; deterministic assertions; LLM-as-a-judge design, judge bias, calibration and human agreement	Build and calibrate a judge against human labels	10
M4	Agent and multi-turn evaluation: trajectory evaluation, tool-call correctness, task completion, conversation quality, regression suites	Trajectory evaluation for the G03 agent	6
M5	Observability: tracing and spans, prompt and version logging, latency and token dashboards, error taxonomies, sampling live traffic, user feedback capture, drift detection	Instrument a live pipeline with end-to-end tracing	6
M6	Cost and performance engineering: token accounting, prompt compression, caching (exact and semantic), model routing and tiering, batching, streaming, budget alarms, unit economics per request	Halve cost per query at constant measured quality	6

Real work scenario — Leadership asks whether a prompt change may ship. Run the regression suite, separate retrieval from generation effects, quantify quality delta with confidence, report cost and latency impact, and issue a go/no-go recommendation with evidence.

M	Topics	Lab	h
M1	AI security: prompt injection (direct and indirect), jailbreaks, data exfiltration through tools and retrieval, RAG corpus poisoning, insecure output handling, supply-chain risk in models and packages, agent permission escalation	Red-team a deployed assistant and document exploit paths	12
M2	Guardrails: input and output filtering, allow/deny policies, refusal design, output validation, escalation thresholds, content classification, defence in depth, fail-safe defaults	Implement layered guardrails and measure bypass rate before and after	10
M3	Privacy and data protection: PII identification and redaction, data minimisation, retention and deletion, consent, purpose limitation, cross-border and residency considerations, Digital Personal Data Protection Act 2023 obligations, audit logging	Build a PII-redaction and retention layer with an audit trail	10
M4	Fairness and transparency: bias sources across data, labels and feedback loops; fairness metrics; disparate impact testing; explainability; model cards and datasheets; documentation duties	Bias audit of a deployed classifier with mitigation proposal	10
M5	Governance and assurance: acceptable-use policy, model risk management, approval workflows, incident response for AI failures, human oversight design, vendor and third-party assessment	Draft an AI acceptable-use and incident-response policy	6

Gate 3 (Week 18): deliver a grounded RAG assistant with an agentic action, layered guardrails, and an evaluation harness reporting retrieval and generation metrics independently, plus a cost-per-query figure. Pass $\geq 70\%$.

8 · Stage D — Role specialisation tracks (Weeks 19–22 · 160 h each)

Each track is a full subject of four modules plus a track project. Learners take exactly one. All tracks assume Gates 1–3 passed.

R1 · AI Engineer

160 h · 48 T / 112 P

Role scope Applies pre-trained models and existing AI tooling to build and operate production AI features; does not train foundation models

Stack Python FastAPI LangChain LangGraph LlamaIndex pgvector Qdrant LiteLLM
Redis Docker RAGAS LangSmith GitHub Actions

M	Topics	Lab	h
M1	Production RAG engineering: query understanding and rewriting, multi-hop and recursive retrieval, reranker selection and tuning, chunk and index experimentation methodology, freshness and invalidation, multi-tenant corpus isolation	Systematic retrieval-tuning study raising recall@k on a fixed corpus with documented method	40
M2	Agentic feature delivery: workflow decomposition, tool contract design, durable state, retries and compensating actions, partial failure handling, human-in-the-loop checkpoints, multi-agent supervision	Multi-step agent with durable state, approval gate and full recovery behaviour	40
M3	Reliability, cost and latency engineering: model routing and fallback chains, provider outage handling, semantic and exact caching, streaming, concurrency and backpressure, rate-limit management, token budgeting, unit economics, SLOs and error budgets	Reduce cost per query by $\geq 50\%$ and p95 latency by $\geq 30\%$ at held quality	40
M4	Ship and operate: service packaging, configuration and secrets, CI/CD with evaluation gates that block regression , canary release and rollback, tracing and alerting, on-call runbooks, post-incident review	Deploy behind an eval-gated pipeline; force a regression and prove the gate blocks it	40

Track project — Enterprise knowledge assistant: RAG over a multi-format corpus with enforced citations, an agentic action with approval gate, layered guardrails, evaluation harness separating retrieval and generation, documented cost and latency budget, deployed via CI/CD with eval gates, tracing and a runbook.

Exit standard Ships a RAG/agent system end to end and defends design under cost, latency and safety constraints; 4+ repositories; interview-ready on RAG and agent system design, evaluation methodology and inference economics

Role scope Adapts, serves and evaluates language models; owns fine-tuning and inference performance

Stack `PyTorch` `Hugging Face Transformers` `Datasets` `PEFT` `TRL` `bitsandbytes` `vLLM`
`TGI` `DeepEval` `Weights & Biases` `MLflow`

M	Topics	Lab	h
M1	Dataset engineering: task definition, instruction dataset construction, synthetic data generation and filtering, quality rubrics, deduplication and decontamination, train/validation/test hygiene, licensing and provenance, annotation operations	Construct, clean and document an instruction dataset with a held-out evaluation split	36
M2	Adaptation techniques: prompt engineering versus RAG versus fine-tuning decision framework; supervised fine-tuning; parameter-efficient methods (LoRA, QLoRA) — rank, alpha, target modules; preference optimisation (DPO); catastrophic forgetting; hyperparameter selection; compute and memory planning	Fine-tune an open-weights model with a parameter-efficient method and version the artefact	44
M3	Inference optimisation and serving: quantisation formats and quality impact, KV-cache mechanics and reuse, paged attention, continuous batching , speculative decoding, tensor parallelism, GPU memory budgeting and utilisation, throughput versus latency tuning, autoscaling and cold start	Serve the tuned model; benchmark tokens/second, p95 latency, memory and cost per million tokens against the baseline	44
M4	Rigorous evaluation and release: task-specific benchmark design, base-versus-tuned comparison, statistical significance, safety and regression evaluation, model cards, staged rollout and rollback criteria	Publish a comparative evaluation report supporting a release decision	36

Track project — Domain-adapted language model: build the dataset, fine-tune with a parameter-efficient method, self-host on an optimised serving stack, and publish a benchmark report versus the base model covering quality, latency, memory and cost, with a documented release recommendation.

Role scope Owns the path to production and the reliability of AI systems in operation

Stack Docker Kubernetes Helm MLflow DVC Airflow Terraform GitHub Actions
Prometheus Grafana Evidently

M	Topics	Lab	h
M1	Containerisation and orchestration: image design and minimisation, multi-stage builds, registries and scanning; Kubernetes objects (pod, deployment, service, ingress, config, secret), resource requests and limits, autoscaling, GPU scheduling, node pools, rollout strategies	Deploy a scalable, resource-bounded model service on Kubernetes with autoscaling	40
M2	Reproducibility and lineage: experiment tracking, model registry and staging transitions, dataset and artefact versioning, deterministic environments, orchestration DAGs, scheduling, backfills, dependency and failure handling	Automated, versioned training-to-registry pipeline with scheduled retraining	40
M3	Continuous delivery for AI: test strategy for ML and LLM systems, evaluation gates in CI/CD , infrastructure as code, environment promotion, canary and blue-green deployment, automated rollback, secrets and supply-chain security	Pipeline that automatically blocks promotion when evaluation scores regress	40
M4	Operations and observability: metrics, logs and traces; data drift, concept drift and prediction drift; performance and cost dashboards; alerting and thresholds; SLO and error-budget management; incident response and post-mortems; capacity planning	Drift detection with alerting plus an incident runbook and simulated incident	40

Track project — Production ML platform: pipeline, registry, evaluation-gated CI/CD, canary deployment with automated rollback, drift monitoring and alerting, cost dashboard, and an operational runbook — demonstrated through an induced failure and recovery.

R4 · Data Scientist

160 h · 56 T / 104 P

Role scope Converts business questions into measured, defensible, decision-grade analysis and models

Stack Python SQL scikit-learn XGBoost statsmodels SHAP Prophet Plotly
Streamlit Power BI/Tableau

M	Topics	Lab	h
M1	Problem framing and metric design: translating business objectives into measurable targets, primary versus guardrail metrics, proxy-metric risk, baselines, cost of errors, decision thresholds, stakeholder alignment	Convert an ambiguous business objective into a measurable analytical plan	36
M2	Inference and experimentation: sampling, estimation, confidence intervals, hypothesis testing, multiple comparisons, power and sample-size calculation, A/B test design and analysis, sequential testing pitfalls, quasi-experiments, confounding and causal reasoning	Design, power and analyse a controlled experiment; write the decision memo	44
M3	Advanced modelling and interpretability: regularisation, ensembles, time-series forecasting and seasonality, survival and uplift concepts, calibration, feature attribution with SHAP, partial dependence, fairness assessment, model limitations	Build a calibrated, explained model with a documented limitations section	44
M4	Communication and decision support: narrative structure, executive summaries, visual encoding principles, dashboard design, uncertainty communication, recommendation framing, handling challenge from stakeholders	Present findings and defend the recommendation to a challenging panel	36

Track project — End-to-end business analytics engagement: framing, data preparation, calibrated model with interpretability, decision dashboard, quantified recommendation with expected impact and uncertainty, delivered as a stakeholder presentation.

Role scope Builds and operates the data and retrieval layer that AI systems depend on

Stack Python SQL Airflow dbt Unstructured pgvector Qdrant OpenSearch
sentence-transformers Great Expectations

M	Topics	Lab	h
M1	Ingestion at scale: source connectors, batch versus streaming, change data capture, incremental and delta sync, complex document parsing (tables, forms, scans), OCR quality, deduplication, idempotent loads, backfill strategy	Production ingestion pipeline over a heterogeneous, frequently changing corpus	40
M2	Transformation and modelling: dimensional modelling, transformation frameworks, testing and data contracts, data quality dimensions, lineage and cataloguing, schema evolution, PII handling in pipelines	Tested transformation layer with enforced data contracts and lineage	40
M3	Retrieval engineering: embedding model selection and benchmarking, chunking experimentation, index type and parameter tuning (HNSW/IVF), metadata filtering, hybrid search fusion, reranking, freshness and TTL, multi-tenancy and access control in retrieval	Tune an index to meet stated recall and p95 latency targets	44
M4	Operating the retrieval layer: retrieval evaluation harness, monitoring recall drift, corpus growth and re-embedding strategy, cost of storage and embedding, capacity planning, disaster recovery, corpus poisoning defence	Retrieval evaluation and monitoring suite with scheduled refresh	36

Track project — Production retrieval platform: ingestion of a multi-format changing corpus, tested transformation layer, tuned hybrid index with reranking, access-controlled multi-tenant retrieval, evaluation harness reporting recall@k, MRR and faithfulness, plus scheduled refresh and monitoring.

Role scope	Designs AI systems that satisfy functional, economic, security and regulatory constraints
Entry	Gates 1–3 at ≥80% plus demonstrated communication competence
Stack	Cloud AI platforms (Bedrock / Azure AI / Vertex AI), Terraform , API gateways, LiteLLM , IAM/KMS

M	Topics	Lab	h
M1	Architecture method: requirements and constraint capture, quality attributes, trade-off analysis, architecture decision records, reference architectures, build-versus-buy, managed versus self-hosted, vendor lock-in and exit strategy	Produce an architecture decision record with a defended trade-off analysis	40
M2	Designing for scale and resilience: capacity planning, throughput modelling, quota and rate-limit architecture, multi-region and failover, graceful degradation, fallback model chains, queueing and asynchronous patterns, disaster recovery objectives	Highly available design operating under provider quota limits, with failure-mode analysis	40
M3	AI economics: cost drivers and modelling, token and inference economics, caching tiers, right-sizing, reserved versus on-demand, storage and egress, total cost of ownership, chargeback and FinOps practice, cost guardrails	Build a defensible cost model and TCO comparison for a stated workload volume	40
M4	Security, compliance and governance architecture: tenancy isolation, network and identity design, encryption and key management, data residency and sovereignty, DPDP Act obligations, auditability, model risk governance, third-party assessment, security review process	Complete a security and governance design review against a control checklist	40

Track project — Enterprise AI reference architecture with a working reference implementation: architecture decision records, scaling and resilience plan, quantified cost model, security and governance review, and an executive presentation defending the design.

Gate 4 (Week 22): track project deployed and demonstrated, with documentation, measured evidence and a technical defence. Pass ≥70%.

9 · Professional & statutory subjects (parallel, Weeks 1–24)

P01 · Professional Practice & Employability

60 h · 24 T / 36 P — statutory DGT/VSQ/N0102

Module	Topics	NOS	h
Introduction to Employability Skills	World of work, employer expectations, workplace norms	—	1.5
Constitutional Values — Citizenship	Rights and duties, civic responsibility, workplace ethics	—	1.5
Becoming a Professional in the 21st Century	Professionalism, accountability, adaptability, lifelong learning	N9001	2.5
Basic English Skills	Workplace reading, writing, speaking and listening; technical English	N9002, N9004	10
Career Development & Goal Setting	Career mapping, goal setting, self-assessment	—	2
Communication Skills	Clear, concise and accurate communication; written reporting; active listening; conflict handling	N9002 PC1–PC3	5
Diversity & Inclusion	Respectful workplaces, inclusive behaviour, bias awareness	N9006, N9014	2.5
Financial & Legal Literacy	Personal finance, contracts, workplace legal basics, confidentiality	N9001 PC5	5
Essential Digital Skills	Productivity tooling, collaboration platforms, digital hygiene, cyber-safety	N9004	10
Entrepreneurship	Opportunity identification, value proposition, basic venture literacy	—	7
Customer Service	Stakeholder and client orientation, expectation management	N9006	5
Getting Ready for Apprenticeship & Jobs	Résumé, portfolio, GitHub profile, mock interviews, offer navigation	—	8

Additionally delivered against generic NOS: establishing and agreeing work requirements with appropriate people; effective time and resource use; correct treatment of confidential information; working within organisational policy (N9001). Integrating work with colleagues, passing on essential information, honouring commitments, raising and solving interpersonal problems (N9002). Agreeing, sourcing, verifying and formatting data and information; rule-based analysis; obtaining guidance where data is problematic (N9004). Building rapport, developing professional relationships and alliances, fostering respect, engaging influential contacts, seeking guidance (N9006).

Real work scenario — You are behind on a committed deliverable because an upstream dependency slipped. Renegotiate scope with the appropriate people in writing, inform affected colleagues, protect confidential client information in the communication, and record the revised commitment.

P02 · Inclusive & Environmentally Sustainable Workplace

30 h · 10 T / 20 P — statutory SSC/N9014

M	Topics & performance criteria	Lab	h
M1	Optimising usage of electricity, energy, materials and water across tasks and activities; energy-aware computing; efficient resource scheduling; measuring consumption	Energy and resource audit of a training/compute workload with reduction plan	8
M2	Segregating recyclable, non-recyclable and hazardous waste for disposal or efficient handling; e-waste policy and practice	Workplace waste segregation and e-waste disposal procedure	6
M3	Understanding the organisation's diversity policy; using internal and external communication appropriately; complying with PwD-inclusive policies for an adaptable and equitable work environment	Review a workplace policy for inclusivity gaps and propose amendments	8
M4	Improving through inclusive recruitment practices, PwD-friendly infrastructure and job design; advocating appropriate verbal and non-verbal communication, schemes and benefits; accessibility in digital products	Accessibility review of a delivered AI interface against inclusion criteria	8

10 · Stage E — Capstone and On-the-Job Training

X01 · Capstone Project

80 h · 10 T / 70 P

Mandatory for certification. Individually owned, industry-scoped, and defended before a panel including at least one external practitioner.

Phase	Activity	Deliverable	h
1 · Scoping	Problem selection, stakeholder definition, success metrics, constraints (latency, cost, safety), risk register	Approved project charter and SRS	10
2 · Design	High-level and low-level design, data flows, interface contracts, evaluation plan defined <i>before</i> building	HLD, LLD and evaluation plan	10
3 · Build	Implementation in sprints with version control, code review and continuous integration	Working system in a reviewed repository	30
4 · Evaluate	Execute the evaluation plan; measure quality, latency, cost; error analysis; iterate on evidence	Evaluation report with baseline comparison	14
5 · Harden	Guardrails, security review, failure handling, observability, documentation, runbook	Hardened deployment and operations documentation	10
6 · Defend	Written report, demonstration, panel defence covering design rationale and trade-offs	Report, demo and defence	6

Capstone assessment rubric (100 marks)

Criterion	Marks	Evidence required
Problem framing & requirements	10	SRS with functional and non-functional requirements; measurable success criteria
Design quality	15	HLD/LLD with rationale, assumptions, constraints, dependencies, interfaces
Implementation correctness	20	Working system; tests passing; coverage evidence
Evaluation rigour	20	Pre-declared metrics; baseline comparison; retrieval and generation separated where applicable; honest error analysis
Engineering & operations	15	Version control discipline, CI, containerisation, observability, runbook
Safety, security & governance	10	Guardrails, PII handling, threat consideration, documented limitations
Communication & defence	10	Report clarity, demonstration, quality of answers under questioning

Integrity rule: unverifiable or unreproducible results score zero on evaluation rigour. Every reported number must be reproducible from the repository by an assessor.

Component	Requirement	h
Induction	Host organisation orientation, tooling access, security and confidentiality briefing, working agreement	6
Supervised delivery	Assigned to a live team backlog; tickets delivered under a named supervisor with code review	60
Operational exposure	Stand-ups, sprint ceremonies, incident observation, release participation, stakeholder interaction	14
Reflection & assessment	Logbook, supervisor evaluation against NOS behaviours, exit review	10

Evidence required: signed OJT logbook with dated entries; supervisor assessment against SSC/N9001, N9002, N9004 and N9006 behaviours; at least three merged contributions or equivalent documented deliverables; exit review record. Where a host placement is unavailable, an approved simulated industry environment must replicate ticket flow, code review, release process and supervision — and must be declared as such.

11 · Real-time work scenario progression

Scenarios escalate from single-task execution to ambiguous, multi-stakeholder ownership. Each is delivered as a timed workplace simulation with realistic artefacts — tickets, incomplete requirements, legacy code, production constraints and stakeholder pressure.

Level	Stage	Scenario type	What is deliberately imperfect	Competency proven
L1 Execute	A	Well-specified ticket with clear acceptance criteria	Nothing — baseline competence	Follows standards, delivers to specification
L2 Diagnose	A–B	Defect report: system behaves incorrectly under specific conditions	Cause unknown; logs incomplete	Reproduces, isolates, fixes, regression-tests
L3 Optimise	B	Working system that violates a stated performance or cost budget	Constraint is binding; naive fixes break accuracy	Profiles, quantifies trade-offs, optimises to budget (N8121)
L4 Build	B–C	Model handed over as a notebook; must become a supported service	No specification, no tests, no error handling	Writes SRS/HLD/LLD, refactors, tests, deploys (N8122)
L5 Integrate	C	Multi-component AI feature spanning retrieval, generation and tools	Sources conflict; upstream data changes mid-build	Systems thinking, interface contracts, evaluation design
L6 Operate	C–D	Live incident: quality regression or outage in production	Under time pressure with partial information	Triage, mitigation, rollback, root cause, post-mortem
L7 Decide	D	Stakeholder requests a change that trades quality against cost or timeline	No correct answer; requires defensible judgement	Evidence-based recommendation and written justification
L8 Own	E	Ambiguous business objective with no defined solution	Requirements must be elicited; scope must be negotiated	End-to-end ownership from ambiguity to defended delivery

11.1 Standing simulation environment

A permanent simulated organisation runs for the full programme duration: a ticket backlog, a code repository with review requirements, a staging and production environment, a rotating on-call roster from Stage C, scheduled releases, injected incidents, and role-played stakeholders (product owner, client, security reviewer, data protection officer). Learners accumulate a verifiable delivery history rather than isolated exercises.

12 · Assessment framework & NOS coverage matrix

12.1 Instruments

Instrument	Frequency	Format	Weight
Module quiz	Per module	Auto-graded objective, concept-bank driven	10%
Coding challenge	Per module	Auto-graded, hidden test cases, time-boxed	10%
Lab portfolio	Continuous	Rubric-graded artefacts in version control	15%
Stage gate	5 gates	Theory paper + extended practical	25%
Workplace simulation	8 scenarios	Observed performance against behavioural rubric	10%
Capstone	Once	Project + report + panel defence	25%
Professional conduct	Continuous	Punctuality, collaboration, code review, communication	5%

12.2 Progression rules

- Every stage gate must be passed at **≥70%** before progression; two re-attempts permitted with mandatory remediation between attempts.
- Statutory subjects **C02 (SSC/N8121)** and **C05 (SSC/N8122)** must each be passed at **≥70% independently**; they cannot be compensated by other marks.
- Overall certification requires **≥70% aggregate**, completion of the capstone, and completion of 90 h OJT with supervisor sign-off.
- Attendance minimum 80% of scheduled practical hours.
- Plagiarism or unreproducible results in any graded artefact voids that assessment.

12.3 NOS coverage matrix

NOS	Taught in	Practised in	Assessed by	Evidence retained
SSC/N8121	C02 (90 h)	G04, R1–R6 projects, capstone	Gate 2 practical (30/70 theory/practical)	Specification sheet, profiling report, optimisation evidence, model card
SSC/N8122	C05 (120 h)	R1–R6 projects, capstone	Gate 2 practical (25/75)	SRS, HLD, LLD, service repository, test suite, defect log, approved PR
SSC/N9001	P01, F08	All sprints and simulations	Continuous + simulation rubric	Sprint records, commitments log, time and resource evidence
SSC/N9002	P01, F02	Pair work, code review, team projects	Peer and trainer rubric	Review comments, collaboration log
SSC/N9004	P01, F03, F05	All reporting and documentation	Documentation rubric	Reports and dashboards in agreed formats
SSC/N9006	P01	Stakeholder simulations, OJT	Observed behaviour + supervisor report	Stakeholder interaction records
SSC/N9014	P02 (30 h)	Lab operations throughout	Practical audit tasks	Resource audit, waste procedure, inclusion review

12.4 Technical concept bank for objective assessment

Question banks are constructed against these concepts, which reflect current hiring-interview expectations: algorithmic complexity and Big-O; parallel architectures (SISD/SIMD/MISD/MIMD); Amdahl's law; profiling and latency percentiles; quantisation trade-offs; KV-cache reuse; continuous batching; GPU utilisation and memory

budgeting; context windows and token accounting; temperature and sampling behaviour; embeddings and semantic similarity; chunking strategy effects; index types and distance metrics; hybrid retrieval and reranking; recall@k, MRR and NDCG; faithfulness and answer relevance; LLM-as-a-judge design and bias; agent tool-call correctness and trajectory evaluation; prompt injection and mitigation; PII redaction and DPDP obligations; drift types; CI evaluation gating; SRS/HLD/LLD content requirements; unit-test design and coverage; defect triage and corrective action.

13 · Laboratory & infrastructure specification

Component	Specification	Purpose
Learner workstation	Minimum 16 GB RAM, 4-core CPU, 256 GB SSD, stable broadband; Linux or Windows with WSL2	Local development across all stages
Shared GPU capacity	Pooled GPU instances with scheduled allocation; sized for fine-tuning and serving workloads	C03, C04, R2 fine-tuning and inference benchmarking
Cloud environment	Per-learner sandboxed cloud accounts with budget caps, least-privilege IAM and automatic teardown	F06, R1–R6 deployment, R3 orchestration
Model access	Managed API access with per-learner keys and spend limits, plus self-hosted open-weights models	G01–G05, R1, R2
Data stores	PostgreSQL with vector extension, a dedicated vector database, object storage, search engine	F05, G02, R5
Platform	Version control with protected branches, CI runners, container registry, experiment tracking, tracing and evaluation platform	All stages; evidence retention
Simulation environment	Ticketing system, staging and production namespaces, incident injection tooling, on-call rota	§11 workplace simulations
Assessment platform	Proctored objective testing, auto-graded coding assessment with hidden tests, rubric capture, evidence archive	§12 assessment and audit
Corpora	Licensed and open document corpora including scanned, tabular and multilingual content; synthetic PII sets for redaction practice	G02, G05, R5
Accessibility	PwD-accessible premises, assistive technology, adjustable workstations, accessible digital materials	SSC/N9014 compliance

Cost control: per-learner spend caps, automatic idle shutdown, quota-based GPU scheduling and cached model responses in graded exercises are mandatory — cloud and inference spend is the dominant variable cost in AI training delivery.

14 · Trainer requirements & delivery standards

Item	Standard
Technical trainer	Graduate in Engineering/Technology/CS or allied, with a minimum of 3 years' industry experience in the subject area; demonstrated production delivery (repository or system evidence)
Statutory-module trainer	Certified against the applicable Trainer QP; familiar with the QP, NOS and assessment criteria
Trainer certification	Trainers delivering NSQF-aligned content must hold current Train-the-Trainer certification as required by the awarding body
Ratio	Maximum 25 learners per trainer for theory; maximum 15 per trainer or assistant for practical; 1:8 for capstone mentoring
Industry mentors	At least one practising industry mentor per track, contributing scenario design and capstone panel participation
Content currency	Curriculum version reviewed and re-validated before every cohort; tech-stack atlas re-confirmed; deprecated tooling replaced with documented change log
Session standard	Every practical session produces a committed artefact; every theory session states its NOS linkage and assessment relevance
Learner support	Structured doubt-clearing hours, remediation pathway after failed gates, mental-health and grievance channel, mentor allocation
Records	Attendance, assessment evidence, artefacts, feedback and OJT records retained per awarding-body audit requirements

15 · Compliance & readiness checklist

#	Action	Owner	Blocking
1	Confirm current live version of QP SSC/Q8113 and Model Curriculum with IT-ITeS SSC; record version numbers	Academic head	Yes
2	Select and register the fresher-appropriate companion QP; document dual-registration rationale	Academic head	Yes
3	Complete NOS-to-module traceability matrix and have it reviewed against SSC assessment criteria	Curriculum lead	Yes
4	Engage an SSC-empanelled assessment agency; agree assessment calendar and evidence formats	Operations	Yes
5	Certify trainers against Trainer QP; record certificates	Operations	Yes
6	Commission lab and cloud infrastructure with spend caps and accessibility provisions	Infrastructure	Yes
7	Apply as FutureSkills Prime content/training partner; submit courses for NSQF approval to unlock learner incentives	Partnerships	No
8	Register as NSDC Training Partner; accredit and affiliate training centres on the SMART portal; pursue star rating	Partnerships	No
9	Secure OJT host organisations or commission the approved simulated industry environment	Placements	Yes
10	Establish evidence-retention, audit and grievance procedures	Quality	Yes
11	Run a pilot cohort; capture completion, assessment and placement data; publish an outcomes report	Programme director	No
12	Schedule per-cohort curriculum re-validation and publish the change log	Curriculum lead	No

16 · Standards referenced & document control

Directly sourced from official documents. The qualification parameters, all National Occupational Standards codes, verbatim performance criteria, knowledge-and-understanding items, generic skills, assessment mark allocations and the 70% pass rule in §2 are read directly from the official National Qualifications Register publication of **QP SSC/Q8113 v2.0**. The statutory hour structure in §3 (144 Theory + 246 Practical + 90 OJT = 480 h), the bridge-module breakdown and the Employability Skills module list are read directly from the corresponding **NSQF Model Curriculum v3.0**.

Verify before the first cohort. QP v2.0 carries review and deactivation dating of 2025 and Model Curriculum v3.0 shows validity to 22/09/2025; the successor documents must be confirmed with the IT-ITeS Sector Skill Council. Model Curriculum v3.0 records the constituent NOS at NSQF Level 5 while the QP header states Level 7 — clarify the applicable level for certification claims.

Industry content. Tooling, techniques and role scoping in §5–§8 reflect the production stack and hiring expectations current as of August 2026 and are expected to drift; §14 mandates re-validation before every cohort.

PRIMARY National Qualifications Register — QP SSC/Q8113 v2.0, AI-Machine Learning Engineer · nqr.gov.in

PRIMARY NSQF Model Curriculum v3.0 — AI-Machine Learning Engineer (480 h structure) · [Model Curriculum PDF](#)

PRIMARY NCVET — National Skills Qualification Framework notification · ncvet.gov.in

PRIMARY Companion qualifications for fresher entry — [Certified AI Associate](#) · [AI Application Developer \(NIELIT, NSQF 4.5\)](#) · [AI-ML Developer](#)

PRIMARY FutureSkills Prime (MeitY–NASSCOM) · futureskillsprime.in — learner incentive structure reported by [DataQuest](#)

PRIMARY NSDC Training Partner registration and SMART centre accreditation · nsdcindia.org · smart.nsdcindia.org

SECONDARY NASSCOM–Deloitte *Advancing India's AI Skills* (demand ~1.25 M by 2027) · deloitte.com

SECONDARY Role scoping and learning-path benchmarks — roadmap.sh AI Engineer · [DeepLearning.AI course catalogue](#)

INDUSTRY 2026 production LLM stack and LLMOps practice (serving, orchestration, retrieval, evaluation layers) · [LLMOps roadmap](#) · [inference engine comparison](#)

Document control

Version	2026.1
Status	For internal approval and awarding-body submission preparation

Prepared for	Anvi Innovate
Review cycle	Before every cohort; mandatory technology re-validation and change log
Next action	Complete §15 blocking items 1–6 and 9–10 prior to first enrolment

Anvi Innovate AI Academy — Complete Curriculum, version 2026.1. Statutory content mapped to QP SSC/Q8113 and NSQF Model Curriculum v3.0 as published on the National Qualifications Register; version currency to be confirmed with the IT-ITeS Sector Skill Council before certification claims are made. Industry content reflects the production stack current as of August 2026 and is subject to scheduled re-validation.